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Artificial Intelligence & Machine Learning

Part - 1

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Long Question

1 1. Data Scientist at a leading digital payment platform such as eSewa, working on large-scale user transaction data. Identify two high-impact business problems where unsupervised learning can provide value .

As a Data Scientist working in digital payment applications like eSewa ,unsupervised learning can play a significant role in solving many business problems by processing large amounts of transaction data, which might be unlabeled in many cases. Two major examples include customer segmentation and fraud detection, which can bring many benefits to the business.

In customer segmentation it allows us to classify customers based on transactional characteristics such as frequency of transactions made, average sum of payments, services most popular among certain groups, etc. Each customer makes transactions differently. Some customers make transactions through eSewa mostly for recharging phones, whereas some use eSewa for utility payments or shopping purposes. Therefore, methods such as K-Means clustering and DBSCAN can detect patterns in user behavior and divide users into groups. The results of these methods can be added to a batch process where customer segments are updated weekly or even monthly. Such information can be used in marketing to provide personalized cashback and loyalty programs. However, one limitation of the method includes selecting the right number of clusters can be challenging and changes in user behavior overtime.

Another significant application of Unsupervised Learning in the eSewa context can be the detection of fraudulent activities. Typically, fraudulent actions are very few and hard to classify, so unsupervised learning is useful .Using anomaly detection techniques such as Autoencoders or Isolation Forest, patterns for ordinary transactions can be learned and then used to detect any abnormality such as unexpected transfers and payments made from unknown locations. Such techniques can be integrated on the transaction system and if any unusual activity is detected actions like OTP authentication, temporary account suspension or security alerts can be triggered. This would help prevent loss of money in addition to ensuring trustworthiness. The disadvantage of such an application could be the existence of errors, especially when valid transactions are mistakenly identified as being fraudulent.

1 In conclusion, unsupervised learning allows eSewa to optimize personalization, security, and decision-making in a business.

Short Question

1. Overfitting is a common challenge in deep learning, especially when models have high capacity.

   It means that the neural network model performs well when it deals with training data but not with unseen data. The reason for this phenomenon is that the model remembers the training dataset and does not generalize the learned patterns. Several ways can be applied to deal with overfitting.

   One of the effective techniques to avoid overfitting is Dropout . In this method, some neurons of the neural network are turned off randomly during training. The model will not concentrate only on specific neurons and will generalize patterns better. However, using this technique will slow down training. It finds practical applications, in image classification when convolutional neural networks utilize dropout to recognize objects correctly in pictures.

   Early stopping is one more technique that can be employed to overcome overfitting. When training the model, it is important to monitor its performance on validation datasets constantly. Once improvement in validation accuracy stops or the validation loss increases, training is stopped. In practice, early stopping is often used in fraud detection tools to avoid overfitting by not allowing the model to learn unnecessary information from the training set.

   Therefore, methods such as dropout and early stopping can be used to improve the accuracy, reliability, and effectiveness of deep learning models.

2. CNN vs RNN

   There are several neural networks models used for various kinds of data in Deep Learning technology. The two models include Convolutional Neural Networks and Recurrent Neural Networks.

   Convolutional Neural Networks are used for image-based data processing. These networks take grid inputs such as images and apply convolution and pooling operations to detect patterns related to edges, shapes, and objects. This process takes place in sequential layers.

Convolutional Neural Networks can share filters between different image parts, hence minimizing the parameters used. This network is mainly used in face recognition, image classification and medical image analysis.

2

RNN models are used for sequential data like text, speech, and time- series data. . RNN processes the input data sequentially while retaining the previously processed data through hidden states. RNN uses the same weights across all time steps. RNNs are often used in machine translation, speech recognition, and stock price prediction.

Training deep learning models face some difficulties during training . Vanishing or exploding gradients occur when gradients become very small or very large, making it difficult to train deep networks. This challenge can be addressed by applying gradient clipping and batch normalization. Overfitting is another challenge that occurs when deep learning models learn from the training set and perform poorly on test sets so techniques like dropout and early stopping help reduce overfitting.